

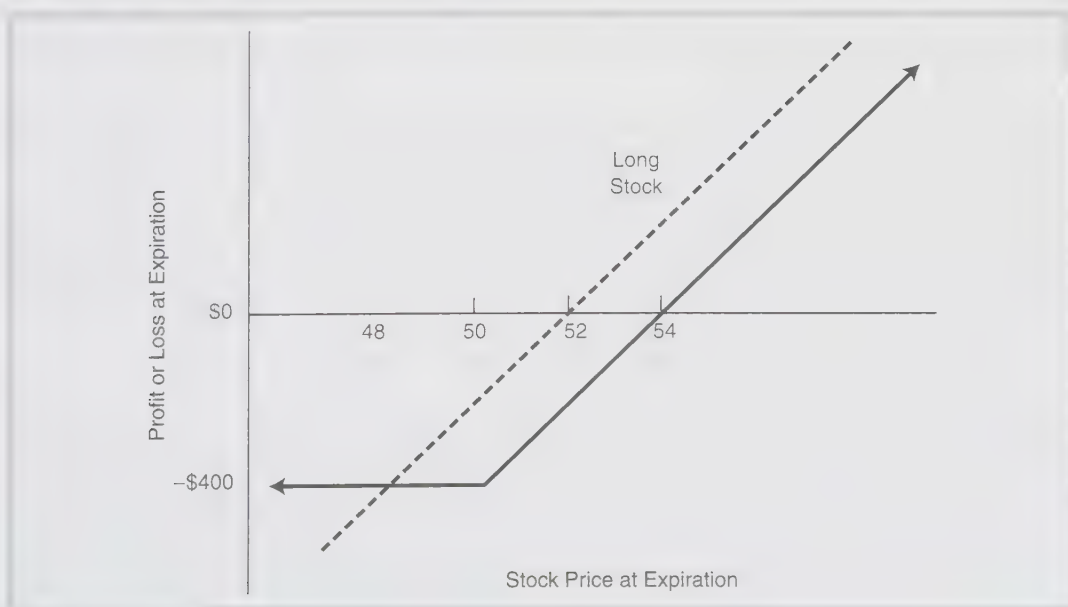
Put Buying in Conjunction with Common Stock Ownership

Another useful feature of put options, in addition to their speculative leverage in a downward move by the underlying stock, is that *the put purchase can be used to limit downside loss in a stock that is owned*. When one simultaneously owns both the common stock and a put on that same stock, he has a position with limited downside risk during the life of the put. This position is also called a synthetic long call, because the profit graph is the same shape as a long call's.

Example: An investor owns XYZ stock, which is at 52, and purchases an XYZ October 50 put for 2. The put gives him the right to *sell* XYZ at 50, so the most that the stockholder can lose on his stock is 2 points. Since he pays 2 points for the put protection, his maximum potential loss until October expiration is 4 points, no matter how far XYZ might decline up until that time. If, on the other hand, the price of the stock should move up by October, the investor would realize any gain in the stock, less the 2 points that he paid for the put protection. *The put functions much like an insurance policy with a finite life*. Table 17-1 and Figure 17-1 depict the results at October expiration for this position: buying the October 50 put for 2 points to protect a holding in XYZ common stock, which is selling at 52. The dashed line on the graph represents the profit potential of the common stock ownership by itself. Notice that if the stock were below 48 in October, the common stock owner would have been better off buying the put. However, with XYZ above 48 at expiration, the put purchase was a burden that cost a small portion of potential profits.

TABLE 17-1.**Results at expiration on a protected stock holding.**

XYZ Price at Expiration	Stock Profit	Put Profit	Total Profit
30	\$2,200	+ \$1,800	- \$ 400
40	- 1,200	+ 800	- 400
50	- 200	- 200	- 400
54	+ 200	- 200	0
60	+ 800	- 200	+ 600
70	+ 1,800	- 200	+ 1,600
80	+ 2,800	- 200	+ 2,600

FIGURE 17-1.**Long common stock and long put.**

This strategy, however, is not necessarily geared to maximizing one's profit potential on the common stock, but rather provides the stock owner with *protection*, eliminating the possibility of any devastating loss on the stock holding during the life of the put. In all the put buying strategies discussed in this chapter and Chapter 15, the put must be paid for in full. That is the only increase in investment.

Although any common stockholder may use this strategy, two general classes of stock owners find it particularly attractive: First, the long-term holder of the stock who is not considering selling the stock may utilize the put protection to limit losses over a short-term horizon. Second, the buyer of common stock who wants some “insurance” in case he is wrong may also find the put protection attractive.

The long-term holder who strongly feels that his stock will drop should probably sell that stock. However, his cost basis may make the capital gains tax on the sale prohibitive. He also may not be entirely sure that the stock will decline—and may want to continue to hold the stock in case it *does* go up. In either case, the purchase of a put will limit the stockholder’s downside risk while still allowing room for upside appreciation. A large number of individual and institutional investors have holdings that they might find difficult to sell for one reason or another. The purchase of a low-cost put can often reduce the negative effects of a bear market on their holdings.

The second general class of put buyers for protection includes the investor who is *establishing* a position in the stock. He might want to buy a put at the same time that he buys the stock, thereby creating a position with profitability as depicted in the previous profit graph. He immediately starts out with a position that has limited downside risk with large potential profits if the stock moves up. In this way, he can feel free to hold the stock during the life of the put without worrying about when to sell it if it should experience a temporary setback. Some fairly aggressive stock traders use this technique because it eliminates the necessity of having to place a stop loss order on the stock. It is often frustrating to see a stock fall and touch off one’s stop loss limit order, only to subsequently rise in price. The stock owner who has a put for protection need not overreact to a downward move. He can afford to sit back and wait during the life of the put, since he has built-in protection.

WHICH PUT TO BUY

The selection of which put the stock owner purchases will determine how much of his profit potential he is giving up and how much risk he is limiting. An out-of-the-money put will cost very little. Therefore, it will be less of a hindrance on profit potential if the underlying stock rises in price. Unfortunately, the put’s protective feature is small until the stock falls to the striking price of the put. Therefore, *the purchase of the out-of-the-money put will not provide as much downside protection as an at- or in-the-money put would*. The purchase of a deeply out-of-the-money put as protection is more like “disaster insurance”: It will prevent a stock owner from experiencing a disaster in terms of a downside loss during the life of the put, but will not provide much protection in the case of a limited stock decline.

Example: XYZ is at 40 and the October 35 put is selling for .50. The purchase of this put as protection for the common stock would not reduce upside potential much at all, only by .50. However, the stock owner could lose 5.50 points if XYZ fell to 35 or below. That is his maximum possible loss, for if XYZ were below 35 at October expiration, he could exercise his put to sell the stock at 35, losing 5 points on the stock, and he would have paid 50 cents for the put, bringing his total loss to 5.50 points.

At the opposite end of the spectrum, the stock owner might buy an in-the-money put as protection. This would quite severely limit his profit potential, since the underlying stock would have to rise above the strike and more for him to make a profit. However, the in-the-money put provides vast quantities of downside protection, limiting his loss to a very small amount.

Example: XYZ is again at 40 and there is an October 45 put selling for 5.50. The stock owner who purchases the October 45 put would have a maximum risk of 50 cents, for he could always exercise the put to sell stock at 45, giving him a 5-point *gain* on the stock, but he paid 5.50 points for the put, thereby giving him an overall maximum loss of 50 cents. He would have difficulty making any profit during the life of the put, however. XYZ would have to rise by more than 5.50 points (the cost of the put) for him to make any total profit on the position by October expiration.

The deep in-the-money put purchase is overly conservative and is usually not a good strategy. On the other hand, it is not wise to purchase a put that is too deeply out-of-the-money as protection. *Generally, one should purchase a slightly out-of-the-money put as protection.* This helps to achieve a balance between the positive feature of protection for the common stock and the negative feature of limiting profits.

The reader may find it interesting to know that he has actually gone through this analysis, back in Chapter 3. Glance again at the profit graph for this strategy of using the put purchase to protect a common stock holding (Figure 17-1). It has exactly the same shape as the profit graph of a simple call purchase. *Therefore, the call purchase and the long put/long stock strategies are equivalent.* Again, by equivalent it is meant that they have similar profit potentials. Obviously, the ownership of a call differs substantially from the ownership of common stock and a put. The stock owner continues to maintain his position for an indefinite period of time, while the call holder does not. Also, the stockholder is forced to pay substantially more for his position than is the call holder, and he also receives dividends whereas the call holder does not. Therefore, “equivalent” does not mean *exactly* the same when comparing call-oriented and put-oriented strategies, but rather denotes that they have similar profit potentials.

In Chapter 3, it was determined that the slightly in-the-money call often offers the best ratio between risk and reward. When the call is slightly in-the-money, the stock is

above the striking price. Similarly, the slightly out-of-the-money put often offers the best ratio between risk and reward for the common stockholder who is buying the put for protection. Again, the stock is slightly above the striking price. Actually, since the two positions are equivalent, the same conclusions should be arrived at; that is why it was stated that the reader has been through this analysis previously.

TAX CONSIDERATIONS

Although tax considerations are covered in detail in a later chapter, an important tax law concerning the purchase of puts against a common stock holding should be mentioned at this time. If the stock owner is already a long-term holder of the stock at the time that he buys the put, the put purchase has no effect on his tax status. Similarly, if the stock buyer buys the stock at the time that he buys the put and identifies the position as a hedge, there is no effect on the tax status of his stock. However, *if one is currently a short-term holder of the common stock at the time that he buys a put, he eliminates any accrued holding period on his common stock. Moreover, the holding period for that stock does not begin again until the put is sold.*

Example: Assume the long-term holding period is 6 months. That is, a stock owner must own the stock for 6 months before it can be considered a long-term capital gain. An investor who bought the stock and held it for 5 months and then purchased a put would wipe out his entire holding period of 5 months. Suppose he then held the put and the stock simultaneously for 6 months, liquidating the put at the end of 6 months. His holding period would start all over again for that common stock. Even though he has owned the stock for 11 months—5 months prior to the put purchase and 6 months more while he simultaneously owned the put—his holding period for tax purposes is considered to be zero!

This law could have important tax ramifications, and one should consult a tax advisor if he is in doubt as to the effect that a put purchase might have on the taxability of his common stock holdings.

PUT BUYING AS PROTECTION FOR THE COVERED CALL WRITER

Since put purchases afford protection to the owner of common stock, some investors naturally feel that the same protective feature could be used to limit their downside risk in the covered call writing strategy. Recall that the covered call writing strategy involves the purchase of stock and the sale of a call option against that stock. The covered write has limited upside profit potential and offers protection to the downside in the amount of

the call premium. The covered writer will make money if the stock falls a little, remains unchanged, or rises by expiration. The covered writer can actually lose money only if the stock falls by more than the call premium received. He has potentially large downside losses. This strategy is known as a *protective collar* or, more simply, a “collar.” (It is also called a “hedge wrapper,” although that is an outdated term.)

The purchase of an out-of-the-money put option can eliminate the risk of large potential losses for the covered write, although the money spent for the put purchase will reduce the overall return from the covered write. One must therefore include the put cost in his initial calculations to determine if it is worthwhile to buy the put.

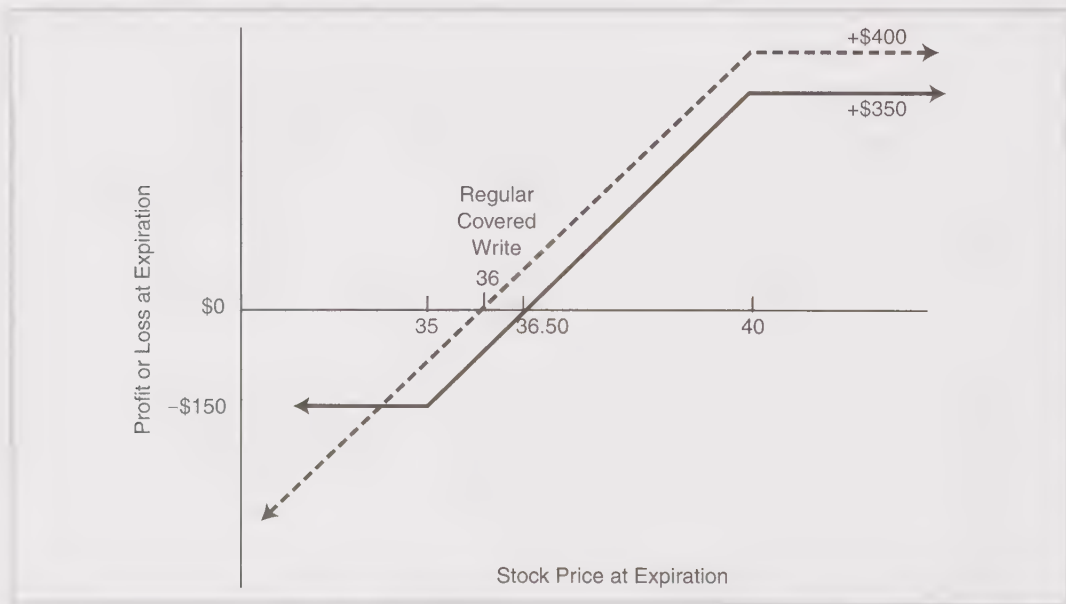
Example: XYZ is at 39 and there is an XYZ October 40 call selling for 3 points and an XYZ October 35 put selling for .50. A covered write could be established by buying the common at 39 and selling the October 40 call for 3. This covered write would have a maximum profit potential of 4 points if XYZ were anywhere above 40 at expiration. The write would lose money if XYZ were anywhere below 36, the break-even point, at October expiration. By also purchasing the October 35 put at the time the covered write is initiated, the covered writer will limit his profit potential slightly, but will also greatly reduce his risk potential. If the put purchase is added to the covered write, the maximum profit potential is reduced to 3.50 points at October expiration. The break-even point moves up to 36.50, and the writer will experience some loss if XYZ is below 36.50 at expiration. However, the most that the writer could lose would be 1.50 points if XYZ were below 35 at expiration. The purchase of the put option produces this loss-limiting effect. Table 17-2 and Figure 17-2 depict the profitability of both the regular covered write and the covered write that is protected by the put purchase.

Commissions should be carefully included in the covered writer's return calculations, as well as the cost of the put. It was demonstrated in Chapter 2 that the covered writer must include all commissions and margin interest expenses as well as all dividends received in order to produce an accurate “total return” picture of the covered write. Figure 17-2 shows that the break-even point is raised slightly and the overall profit potential is reduced by the purchase of the put. However, *the maximum risk is quite small and the writer need never be forced to roll down in a disadvantageous situation.*

Recall that the covered writer who does not have the protective put in place is forced to roll down in order to gain increased downside protection. Rolling down merely means that he buys back the call that is currently written and writes another call, with a lower striking price, in its place. This rolling-down action can be helpful if the stock stabilizes after falling; but if the stock reverses and climbs upward in price again, the covered writer who rolled down would have limited his gains. In fact, he may even have “locked in” a loss. The writer who has the protective put need not be bothered with such things. He never has to roll down, for he has a limited maximum loss. Therefore, he should never get into

TABLE 17-2.**Comparison of regular and protected covered writes.**

XYZ Price at Expiration	Stock Profit	October 40 Call Profit	October 35 Put Profit	Total Profit
25	-\$1,400	+\$300	+\$950	-\$150
30	- 900	+ 300	+ 450	- 150
35	- 400	+ 300	- 50	- 150
36.50	- 250	+ 300	- 50	0
38	- 100	+ 300	- 50	+ 150
40	+ 100	+ 300	- 50	+ 350
45	+ 600	- 200	- 50	+ 350
50	+ 1,100	- 700	- 50	+ 350

FIGURE 17-2.**Covered call write protected by a put purchase.**

a “locked-in” loss situation. This can be a great advantage, especially from an emotional viewpoint, because the writer is never forced to make a decision as to the future price of the stock in the middle of the stock’s decline. With the put in place, he can feel free to take no action at all, since his overall loss is limited. If the stock should rally upward later, he will still be in a position to make his maximum profit.

The longer-term effects of buying puts in combination with covered writes are not easily definable, but it would appear that the writer reduces his overall rate of return slightly by buying the puts. This is because he gives something away if the stock falls slightly, remains unchanged, or rises in price. He only “gains” something if the stock falls heavily. Since the odds of a stock falling heavily are small in comparison to the other events (falling slightly, remaining unchanged, or rising), the writer will be gaining something in only a small percentage of cases. However, the put buying strategy may still prove useful in that it removes the emotional uncertainty of large losses. The covered writer who buys puts may often find it easier to operate in a more rational manner when he has the protective put in place.

This strategy is equivalent to one that has been described before, the bull spread. Notice that the profit graph in Figure 17-2 has the same shape as the bull spread profit graph (Figure 7-1). This means that the two strategies are equivalent. In fact, in Chapter 7 it was pointed out that the bull spread could sometimes be considered a “substitute” for covered writing. Actually, the bull spread is more akin to this strategy—the covered write protected by a put purchase. There are, of course, differences between the strategies. They are equivalent in profit and loss potential, but the covered writer could never lose all his investment in a short period of time, although the spreader could. In order to actually use bull spreads as substitutes for covered writes, one would invest only a small portion of his available funds in the spread and would place the remainder of his funds in fixed-income securities. That strategy was discussed in more depth in Chapter 7.

NO-COST COLLARS

The “collar” strategy is often arrived at in another manner: a stockholder begins to worry about the downside potential of the stock market and decides to buy puts on his stock as protection. However, he is dismayed by the cost of the puts and so he *also* considers the sale of calls. If he buys an out-of-the-money put, it is quite possible that he might be able to sell an out-of-the-money call whose proceeds completely cover the cost of the put. Thus, he has established a protective *collar* at no cost—at least no debit. His “cost” is the fact that he has forsaken the upside profit potential on his stock, above the striking price of the written call.

In fact, certain large institutional traders are able to transact collars through large over-the-counter option brokers, such as Goldman Sachs or Morgan Stanley. They might even give the broker instructions such as this: “I own XYZ and I want to buy a put 10 percent out of the money that expires in a year. What would the striking price of a one-year call have to be in order to create a no-cost collar?” The broker might then tell him that such a call would have to be struck 30 percent out of the money. The actual strike

TABLE 17-3.

Highest Call Strike That Pays for an At-the-Money Put (Assuming 2.5 years to expiration)

Volatility	Call Strike of Underlying
30%	30% out of money
40%	35% out of money
50%	40% out of money
70%	50% out of money
100%	70% out of money

price of the call would depend on the volatility estimate for the underlying stock, as well as interest rates and dividends. These types of transactions occur with a fair amount of frequency.

Some very interesting situations can be created with long-term options. One of the most interesting occurred in 1999, when a company that owned 5 million shares of Cisco (CSCO) decided it would like to hedge them by creating a no-cost collar over the next three years. At the time, CSCO was trading at about 130, and its volatility was about 50%. It turns out that a three-year put struck at 130 sells for about the same price as a three-year call struck at 200! That may seem illogical, but the figures can be checked out with the aid of an option-pricing model. Thus, this company was able to hedge all of its CSCO stock, with no downside risk (the striking price of the puts was the same as the current stock price) and still had profit potential of over 50% to the upside over the next three years.

Thus, one should consider using LEAPS options when he establishes a collar—even if he is not an institutional trader—because the striking price of the calls can be quite high in comparison to that of the put's strike or in comparison to the price of the underlying stock. Table 17-3 shows how far out-of-the-money a written call could be that still covers the cost of buying an at-the-money put. The time to expiration in this table is 2.5 years—the longest term listed option that currently exists.

USING LOWER STRIKES AS A PARTIAL COVERED WRITE

It should also be pointed out that one does not necessarily have to forsake all of the profit potential from his stock. He might buy the puts, as usual, and then sell calls with a somewhat lower strike than needed for a low-cost collar, but the quantity of calls sold would be less than that of stock owned. In that way, there would be unlimited profit potential on *some* of the shares of the underlying stock.

Example: Suppose that the following prices exist:

XYZ: 61

Apr 55 put: 1

Apr 65 call: 2

Furthermore, suppose that one owns 1000 shares of XYZ. Thus, the purchase of 10 Apr 55 puts at 1 point apiece would protect the downside. In order to cover the cost of those puts (\$1000), one need only sell *five* of the Apr 65 calls at 2 points apiece. Thus, the protection would have cost nothing and there would still be unlimited profit potential on 500 of the shares of XYZ, since only five calls were sold against the 1000 shares that are owned.

In this manner, one could get quite creative in constructing collars—deciding what call strike to use in order to strike a balance between paying for the puts and allowing upside profit potential. The lower the strike he uses for the written calls, the fewer calls he will have to write; the higher the strike of the written calls, the more calls will be necessary to cover the cost of the purchased puts. The tradeoff is that a lower call strike allows for more eventual upside profit potential, but it limits what has been written against to a lower price.

Using the above example once again, these facts can be demonstrated:

Example (continued): As before, the same prices exist, but now one more call will be brought into the picture:

XYZ: 61

Apr 55 put: 1

Apr 65 call: 2

Apr 70 call: 1

As before one could sell *five* of the Apr 65 calls to cover the cost of ten puts, or as an alternative he could sell *ten* of the Apr 70 calls. If he sells the five, he has unlimited profit potential on 500 shares, but the other 500 shares will be called away at 65. In the alternative strategy, he has limited upside profit potential, but nothing will be called away until the stock reaches 70. Which is “better?” It’s not easy to say. In the former strategy, if the stock climbs all the way to 75, it results in the same profit as if the stock is called away at 70 in the latter strategy. This is true because 500 shares would be worth 75, but the other 500 would have been called away at 65—making for an average of 70. Hence, the former strategy only outperforms the latter if the stock actually climbs *above* 75—a rather unlikely event, one

would have to surmise. Still, many investors prefer the former strategy because it gives them protection without asking them to surrender all of their upside profit potential.

ADJUSTING THE COLLAR

A collar might be adjusted if the underlying stock declines sharply in price. After the stock has dropped, the put would be worth a considerable amount, while the call would be worth very little. If the investor felt that the majority of the decline in the stock was finished, he could merely sell the put. Whether or not he covered the call, that would leave him with large profit potential if the stock should rally. On the other hand, if the investor is not certain that the stock has stopped declining, he might roll just the put—or perhaps both the put and the call—down to lower strikes, thereby taking a large credit out of the position in doing so (the credit comes from selling the original put, which is now quite valuable). As a third choice, he could also consider selling some out-of-the-money puts against the puts that are owned. This would bring in some credit, but would expose the stock to losses below the striking price of the short puts.

On the other hand, if the underlying stock increases substantially in price after the collar has been established, the only way out of the collar is to cover the written calls—and that is going to require a (large) debit. Of course, the underlying stock has risen in price, so that is an unrealized profit that could be used to offset the loss in the calls. In essence, there is no convenient exit strategy from a collar on the upside.

In summary, one can often be quite creative with the “collar” strategy. One thing to keep in mind: if one sells options against stock that he has no intention of selling, he is actually writing *naked* calls in his own mind. That is, if one owns stock that “can’t” be sold—perhaps the capital gains would be devastating or the stock has been “in the family” for a long time—then he should not sell covered calls against it, because he will be forced into treating the calls as naked (if he refuses to sell the stock). This can cause quite a bit of consternation if the underlying stock rises significantly in price—consternation that could have easily been avoided by not writing calls against the stock in the first place.